**D. Y. PATIL COLLEGE OF ENGINEERING AKURDI, PUNE – 411044**

**A. Y. 2026 - 27**

**SUBJECT: 410246: Laboratory Practice III**

**LAB INDEX** **BE Div.: A, B, C**

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|---|---|---|---|---|
| **Group A: Design and Analysis of Algorithms** | | | | |
| 1 | | Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity. | | |
| 2 | | Write a program to implement Huffman Encoding using a greedy strategy | | |
| 3 | | Write a program to solve a fractional Knapsack problem using a greedy method. | | |
| 4 | | Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy. | | |
| 5 | | Design n-Queens matrix having first Queen placed. Use backtracking to place remaining Queens to generate the final n-queen's matrix. | | |
| 6 | | **Mini Project**<br>Implement single-threaded and multithreaded matrix multiplication and compare their performance.<br>**OR**<br>Implement merge sort and multithreaded merge sort, comparing execution time and best- and worst-case performance.<br>**OR**<br>Implement and compare Naive and Rabin-Karp string matching algorithms using the same input.<br>**OR**<br>Implement and compare exact and approximation algorithms for the Travelling Salesperson Problem. | | |
| **Group B: Machine Learning** | | | | |
| 1 | | Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks:<br>1. Pre-process the dataset. 2. Identify outliers.<br>3. Check the correlation. 4. Implement linear regression and random forest regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc. | | |
| 2 | | Classify the email using the binary classification method. Email Spam detection has two states: a) Normal State – Not Spam, b) Abnormal State – Spam. Use K-Nearest Neighbors and Support Vector Machine for classification. | | |
| 3 | | Given a bank customer, build a neural network-based classifier that can determine whether they will leave or not in the next 6 months. | | |
| 4 | | Implement Gradient Descent Algorithm to find the local minima of a function. | | |
| 5 | | Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset. Determine the number of clusters using the elbow method. | | |

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| 6 | | **Mini Project**<br>Analyze Indian stock market trends (2000–2020) and predict future stock returns using machine learning.<br>**OR**<br>Predict Titanic passenger survival using machine learning and passenger data.<br>**OR**<br>Develop a signature identification application using a self-created dataset of college students' signatures. | | |
| **Group C: Blockchain Technology** | | | | |
| 1 | | Installation of MetaMask and study spending Ether per transaction | | |
| 2 | | Create your own wallet using Metamask for crypto transactions. | | |
| 3 | | Write a smart contract on a test network, for Bank account of a customer for following operations:<br>• Deposit money • Withdraw Money • Show balance | | |
| 4 | | Write a program in solidity to create Student data. Use the following constructs: • Structures • Arrays • Fallback<br>Deploy this as smart contract on Ethereum and Observe the transaction fee and Gas values. | | |
| 5 | | Write a survey report on types of Blockchains and its real time use cases. | | |
| 6 | | **Mini Project**<br>Develop a blockchain-based decentralized e-voting application (dApp)<br>**OR**<br>Develop a Blockchain based application for transparent and genuine charity<br>**OR**<br>Develop a Blockchain based application for health related medical records<br>**OR**<br>Develop a Blockchain based application for mental health | | |

# CERTIFICATE

This is Certified that Mr./Ms._____ Roll No: _____of Class BE Computer [A/B/C] has carried out the above practical/termwork with in the four walls of the college. His/Her performance is satisfactory & attendance is _____%

Date: Faculty Member in-charge Head of the Department Principal